**System Hardening Report**
**Kali Linux VM (lab-vm)**

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Date: 03/11/2026

### 1. Introduction

This report documents the hardening of a Kali GNU/Linux Rolling 2026.3 virtual machine. It covers six controls: patching, least-privilege users, file permissions, firewall, service reduction, and audit verification. For each control, the report shows the before state, the after state, and how it reduces security risk.

### 2. Summary Table.

| Control | Before | After | Security impact |
|---|---|---|---|
| Patching | 1449 packages were upgradable | All packages upgraded | Closes known vulnerabilities |
| Least privilege | No role-based users | alice, bob, charlie, david in separate groups; only kali has sudo | Limits damage if one account is hacked |
| File permissions | Files readable by everyone (-rw-r--r--) | Mode 640, owned by root:security and root:hr | Only the right group can read sensitive files |
| Firewall | UFW inactive | Deny incoming, allow outgoing, only SSH (22) allowed | Blocks unwanted connections |
| Services | Apache running, port 80 open | Apache disabled, port 80 closed | Smaller attack surface |
| Audit | Not verified | All checks PASS | Proves the hardening worked |

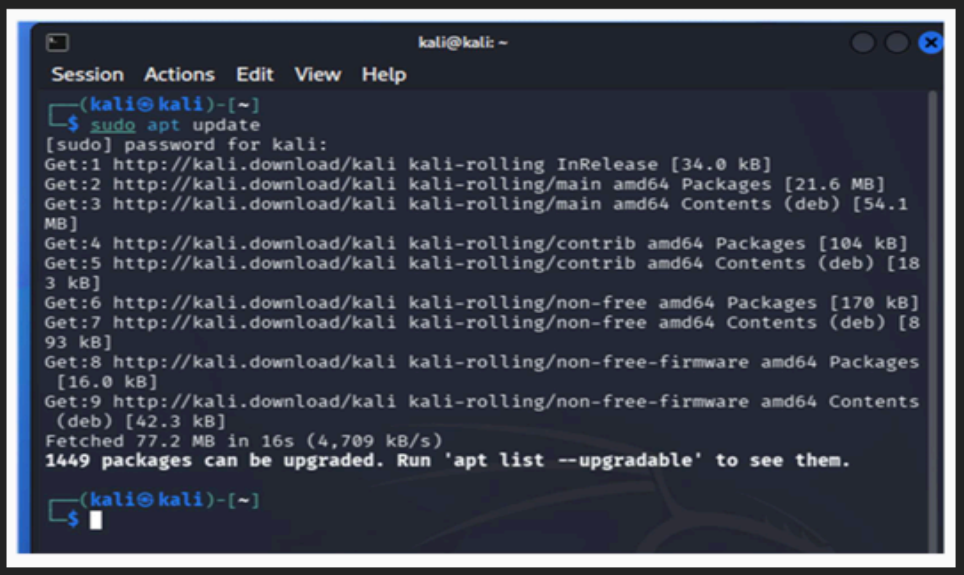
**3. Control Details.**

### 1.1 Q1: Patching

**BEFORE**

**Command: sudo apt update**

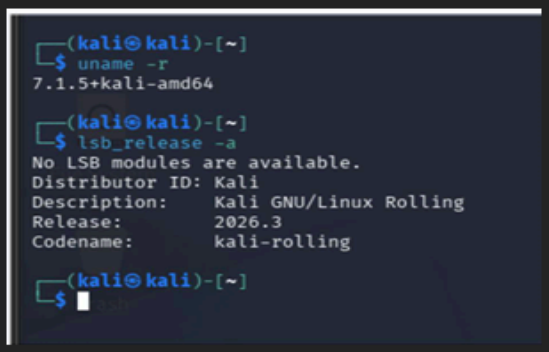
*Pick the screenshot that says '1449 packages can be upgraded'.*




**AFTER**

**Command: uname -r ; lsb_release -a**

***Shows kernel 7.1.5+kali-amd64 and Kali Rolling 2026.3.***




- **What changed: ran `sudo apt update` and `sudo apt upgrade -y`.**
- **Why it matters: updates fix known security holes in installed software, so attackers cannot use old bugs.**

## 1.2 Q2: Least-privilege users

**BEFORE**

**Command: id alice**
*Shows alice has only her own group.*

```
┌──(kali㉿kali)-[~]
└─$ id alice
uid=1001(alice) gid=1004(alice) groups=1004(alice)
```

**AFTER**

**Command: groups alice / bob / charlie / david**
*Shows each user in the right group (security, development, hr).*

```
┌──(kali㉿kali)-[~]
└─$ sudo  -l -U charlie
[sudo] password for kali:
User charlie is not allowed to run sudo on kali.

┌──(kali㉿kali)-[~]
└─$ sudo  -l -U alice
User alice is not allowed to run sudo on kali.

┌──(kali㉿kali)-[~]
└─$ sudo  -l -U bob
User bob is not allowed to run sudo on kali.

┌──(kali㉿kali)-[~]
└─$ sudo  -l -U david
User david is not allowed to run sudo on kali.

┌──(kali㉿kali)-[~]
└─$
```

- **What changed: created groups security, development and hr, created users alice, bob, charlie and david, and added each to a group using `groupadd`, `useradd` and `usermod -aG`.**
- **Why it matters: each person gets only the access they need. If one account is hacked, the attacker cannot reach everything.**

## 1.3 Q3: File permissions

**BEFORE**

**Command: ls -l /company/sensitive.conf**

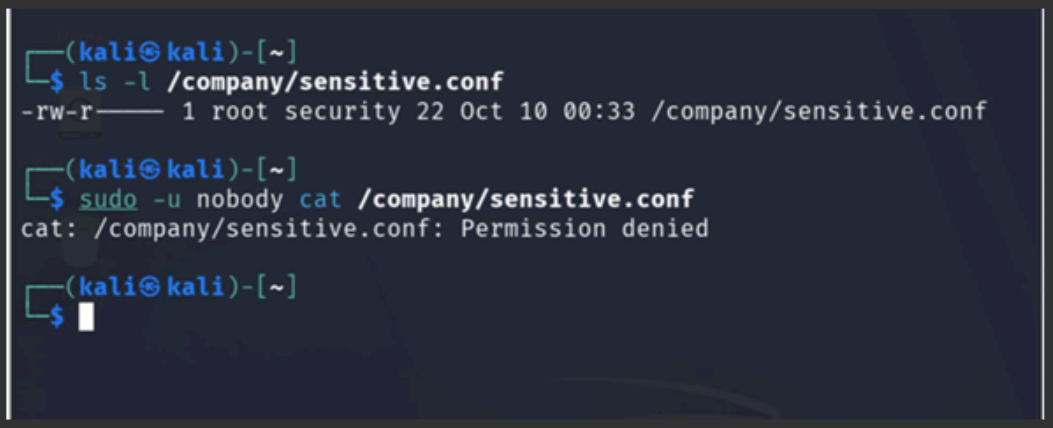
*Shows -rw-r--r-- root root (everyone can read).*




**AFTER**

**Command: ls -l /company/sensitive.conf ; sudo -u nobody cat ...**

*Shows -rw-r----- root security and 'Permission denied'.*




- **What changed: used `chown root:security` and `chmod 640` on sensitive.conf, and `chown root:hr` and `chmod 640` on salary.txt.**
- **Why it matters: only the owner and the group can read the file. Other users get "Permission denied".**

## 1.4 Q4: Firewall

**BEFORE**

**Command: sudo ufw status**

```
┌──(kali㉿kali)-[~]
└─$ sudo ufw status
Status: inactive
```

**AFTER**

**Command: sudo ufw status verbose**

*Shows Status: active, deny (incoming), 22/tcp ALLOW IN.*

```
┌──(kali㉿kali)-[~]
└─$ sudo ufw enable
Firewall is active and enabled on system startup
```

- **What changed: ran `sudo ufw default deny incoming`, `sudo ufw default allow outgoing`, `sudo ufw allow 22/tcp`, and `sudo ufw enable`.**
- **Why it matters: all incoming traffic is blocked except SSH, so attackers cannot reach other services.**

## 1.5 Q5: Services and ports

**BEFORE**

**Command: nmap -sT -p- 127.0.0.1**

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl start apache2

┌──(kali㉿kali)-[~]
└─$ nmap -sT -p- 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-03 09:17 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00014s latency).
Not shown: 65534 closed tcp ports (conn-refused)
PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 2.58 seconds
```

**AFTER**

**Command: sudo systemctl disable --now apache2 ; nmap -sT -p- 127.0.0.1**

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl disable --now apache2

┌──(kali㉿kali)-[~]
└─$ nmap -sT -p- 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-03 09:20 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00016s latency).
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 3.65 seconds
```

- **What changed: scanned with `nmap`, and ran `sudo systemctl disable --now apache2`.**
- **Why it matters: a service that is not running cannot be attacked. Fewer open ports means a smaller attack surface.**

## 1.6 Q6: Audit

**BEFORE**

**Command: apt list --upgradable**
*The 4 upgradable packages seen at the end of Q1.*

```
┌──(kali㉿kali)-[~]
└─$ apt list --upgradable

┌──(kali㉿kali)-[~]
└─$
```

**AFTER**

**Command: Audit commands (Steps 1 to 5)**

```
┌──(kali㉿kali)-[~]
└─$ getent group sudo
sudo:x:27:kali
```

Step 03:

```
┌──(kali㉿kali)-[~]
└─$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
```

Step 04:

```
┌──(kali㉿kali)-[~]
└─$ nmap -sT -p- 127.0.0.1
Starting Nmap 7.991 ( https://nmap.org ) at 2026-11-03 08:17 -0500
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 2.10 seconds
```

Step 05:

```
┌──(kali㉿kali)-[~]
└─$ ls -l /etc/passwd /etc/shadow
-rw-r--r-- 1 root root   3412 Oct  6 00:22 /etc/passwd
-rw-r----- 1 root shadow 1860 Oct  6 00:34 /etc/shadow
```

- **What changed: ran `apt list --upgradable`, `cat /etc/passwd ; groups`, `getent group sudo`, `sudo ufw status verbose`, `nmap`, and `ls -l /etc/passwd /etc/shadow`.**
- **Why it matters: this proves the hardening is really working and records it in a checklist.**

**4. Overall Impact**

The VM moved from an exposed default state to a hardened baseline. Patching closed known vulnerabilities. Role-based users and restricted file permissions limit who can see sensitive data. The firewall allows only SSH inbound, and disabling unused services reduced the open ports. The audit confirmed that every control is working.

**5. Submission Checklist**

**[✓] All six controls documented with before/after**
**[✓] Screenshots attached for each control**
**[✓] Security impact explained per control**
**[✓] Report + GitHub link uploaded to ZOSIMA**